Hafzan QT Research Terminal

Glossary of Terms

Derivatives Analytics v4.2 • Quantitative Finance Reference

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This document provides a comprehensive reference for every term, model, and mathematical concept used in the Hafzan QT Research Terminal. Content is organised by panel in the order they appear in the application. All mathematical notation follows standard quantitative finance conventions.

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Topbar & General Market Data

SPX The S&P 500 Index. A market-capitalisation-weighted index of 500 large-cap US equities, maintained by S&P Dow Jones Indices. The pre-eminent global benchmark for equity market performance. Used by practitioners as the reference underlying for equity derivatives, portfolio attribution, and risk-factor models.

VIX The CBOE Volatility Index. A real-time measure of the market's 30-day forward implied volatility on the S&P 500, derived from the prices of SPX options across a wide range of strikes. Formally defined as:

$$\text{VIX} = 100 \times \sqrt{\frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2}$$

where $Q(K_i)$ is the midpoint of the bid-ask spread for each option with strike K_i, F is the forward index level, and K_0 is the first strike below F. Colloquially called the "fear gauge." Exhibits mean reversion and spikes sharply during equity market sell-offs.

10Y The 10-Year US Treasury yield. The annualised return demanded by investors on 10-year US government bonds. Treated as the global risk-free rate benchmark. Drives the discount factor in derivatives pricing, affects the carry of leveraged positions, and serves as the reference rate for Rho sensitivity in options models.

DXY The US Dollar Index. A geometrically weighted index measuring the dollar against a basket of six currencies: EUR (57.6%), JPY (13.6%), GBP (11.9%), CAD (9.1%), SEK (4.2%), CHF (3.6%). A rising DXY indicates dollar strength, which typically pressures USD-denominated commodities (oil, gold) and EM assets.

UTC Coordinated Universal Time. The primary time standard by which the world regulates clocks and time. Used in all financial markets for timestamping trades and data. The terminal clock displays the current UTC time updated in real-time.

Panel 01 — Implied Volatility Surface

Core Concepts

Implied Volatility (IV) The value of σ that, when substituted into a pricing model (here Black-Scholes-Merton), causes the model price to equal the observed market price of an option. IV is a forward-looking, market-implied quantity and cannot be observed directly. Formally, it is the solution $\hat{\sigma}$ to:

$$C_{\text{market}}(S, K, T, r, q) = C_{\text{BSM}}(S, K, T, r, q, \hat{\sigma})$$

IV encodes the market's collective expectation of future realised volatility for a given strike and expiry.

Volatility Surface A three-dimensional mapping $\sigma_{\text{imp}}(K, T)$ of implied volatility as a function of strike K and time to expiry T. Empirically, the surface is neither flat nor smooth: it exhibits

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a "smile" or "skew" in the strike dimension and a term structure in the expiry dimension. The surface must be arbitrage-free: it must be consistent with a non-negative, integrable risk-neutral density.

Strike Price (K) The contractually fixed price at which the holder of an option has the right to transact the underlying asset. For calls, the holder pays K to receive the asset. For puts, the holder receives K in exchange for delivering the asset.

Time to Expiry (T) The remaining lifetime of an options contract expressed in years. $T = 0$ at expiry, $T = 0.25$ is a 3-month option, $T = 1$ is a 1-year option. Appears under the square root in the BSM formula, reflecting that uncertainty compounds with the square root of time.

Moneyness A dimensionless characterisation of a strike relative to the current underlying price. An *at-the-money* (ATM) option has $S \approx K$. An *in-the-money* (ITM) call has $S > K$. An *out-of-the-money* (OTM) call has $S < K$.

Log-Moneyness (k) Defined as $k = \ln(K/F)$ where $F = Se^{(r-q)T}$ is the forward price. A scale-invariant, symmetric measure of moneyness preferred in parametric models. ATM corresponds to $k = 0$. OTM puts have $k < 0$; OTM calls have $k > 0$.

Volatility Smile The empirical pattern in which implied volatility, plotted against strike for a fixed expiry, forms a U-shaped curve. Both deep OTM puts and deep OTM calls trade at higher IV than ATM options. In equity markets the smile is asymmetric, with puts trading at a premium to calls — this asymmetric form is called the *skew*.

Surface Parameters

ATM Volatility The implied volatility of the option whose strike equals the current forward price. It anchors the entire surface and corresponds to $\sigma_{\text{imp}}(K = F, T)$. It is the most liquid point on the surface and the reference for quoting vol in many markets.

Put Skew The slope of the implied volatility curve in the strike dimension, measured at the ATM point. In equity markets, put skew is negative: as K decreases (OTM puts), IV increases. This reflects the greater market demand for downside protection relative to upside participation. Quantified as $\partial\sigma_{\text{imp}}/\partial k\big|_{k=0}$.

Curvature The second-order term in the smile. Controls how quickly IV rises in both wings away from ATM. High curvature produces a pronounced smile shape. Proportional to $\partial^2\sigma_{\text{imp}}/\partial k^2\big|_{k=0}$. Related to the excess kurtosis of the risk-neutral distribution.

Term Structure The variation of ATM implied volatility as a function of expiry T, holding $k = 0$. The term structure $\sigma_{\text{ATM}}(T)$ reflects the market's expectation of how vol will evolve over time. It can be in contango ($\sigma_{\text{short}} < \sigma_{\text{long}}$) or backwardation ($\sigma_{\text{short}} > \sigma_{\text{long}}$), and steepens around scheduled events.

Vol-of-Vol (ν) Volatility of volatility. The parameter controlling how much the instantaneous volatility itself fluctuates. High vol-of-vol inflates the wings of the smile and is the primary driver of smile curvature in stochastic volatility models. A core parameter in SABR and Heston.

ATM Slope The gradient of the volatility surface at the ATM point in the log-moneyness direction. A steep negative slope indicates strong put skew. Directly related to the correlation ρ

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between the asset price and its volatility process in stochastic vol models.

1M/1Y Term Ratio The ratio of 1-month ATM implied volatility to 1-year ATM implied volatility: $\sigma_{\text{ATM}}(T = 1/12)/\sigma_{\text{ATM}}(T = 1)$. A ratio above 1 indicates the term structure is in backwardation (short-dated vol elevated). A ratio below 1 indicates contango (long-dated vol elevated, typical of calm markets).

Parametric Models

SVI — Stochastic Volatility Inspired Introduced by Jim Gatheral (2004). A parametric model for the total implied variance $w(k) = \sigma^2_{\text{imp}}(k) \cdot T$ as a function of log-moneyness:

$$w(k) = a + b\left[\rho(k-m) + \sqrt{(k-m)^2 + s^2}\right]$$

The five parameters $\{a, b, \rho, m, s\}$ control the ATM variance level, the slope, the asymmetry, the shift, and the smoothness of the smile respectively. Under certain parameter constraints, SVI is free of static arbitrage and can represent any smile consistent with a non-negative risk-neutral density.

SABR — Stochastic Alpha Beta Rho A stochastic volatility model introduced by Hagan, Kumar, Lesniewski & Woodward (2002). The asset price F and its volatility α follow coupled stochastic differential equations:

$$\mathrm{d}F = \alpha F^{\beta}\,\mathrm{d}W_1, \qquad \mathrm{d}\alpha = \nu\alpha\,\mathrm{d}W_2, \qquad \mathrm{d}W_1\,\mathrm{d}W_2 = \rho\,\mathrm{d}t$$

The key feature is an approximate closed-form implied volatility formula, making it fast to calibrate. Parameters: α (ATM vol), β (CEV exponent, often set to 0.5 or 1), ρ (spot-vol correlation driving skew), ν (vol-of-vol driving curvature). Widely used in interest rate derivatives markets.

Heston Model A stochastic volatility model introduced by Steven Heston (1993). The asset price S and instantaneous variance v follow:

$$\mathrm{d}S = \mu S\,\mathrm{d}t + \sqrt{v}\,S\,\mathrm{d}W_1$$

$$\mathrm{d}v = \kappa(\theta - v)\,\mathrm{d}t + \xi\sqrt{v}\,\mathrm{d}W_2$$

where κ is the mean-reversion speed, θ is the long-run variance, and ξ is the vol-of-vol. The two Brownian motions are correlated: $\mathrm{d}W_1\,\mathrm{d}W_2 = \rho\,\mathrm{d}t$. Heston (1993) derived a semi-analytical option pricing formula via characteristic functions, making it computationally tractable. It captures the term structure of volatility better than BSM and produces a persistent smile.

Panel 02 — Options Greeks Engine

The Black-Scholes-Merton Framework

Black-Scholes-Merton Model (BSM) The foundational European options pricing model derived independently by Fischer Black & Myron Scholes (1973) and Robert Merton (1973). Assumes the underlying asset follows GBM under the risk-neutral measure, constant volatility, continuous trading, no transaction costs, and a constant risk-free rate. The model with continuous dividend yield q gives:

$$C = Se^{-qT}\mathcal{N}(d_1) - Ke^{-rT}\mathcal{N}(d_2)$$

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$$P = Ke^{-rT}\mathcal{N}(-d_2) - Se^{-qT}\mathcal{N}(-d_1)$$

d_1 and d_2 Intermediate values in the BSM formula:

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

d_2 is the risk-neutral probability that the option expires in the money: $\mathbb{P}^Q(S_T > K) = \mathcal{N}(d_2)$. d_1 is the Δ-adjusted probability, reflecting the expected asset price conditional on finishing in the money.

$\mathcal{N}(x)$ — normCDF The cumulative distribution function of the standard normal distribution:

$$\mathcal{N}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}}\, \mathrm{d}t$$

Implemented in the terminal using the Abramowitz & Stegun rational polynomial approximation, which achieves a maximum absolute error of less than 1.5×10^{-7} without numerical integration.

$\mathcal{N}'(x)$ — normPDF The probability density function of the standard normal distribution:

$$\mathcal{N}'(x) = \phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Appears in the formulae for Gamma, Vega, and Theta. Represents the marginal probability density at x.

Continuous Dividend Yield (q) An annualised rate at which the underlying continuously pays dividends. Reduces the effective cost of carrying the asset. Modifies BSM by discounting the spot price: the "dividend-adjusted forward" is $F = Se^{(r-q)T}$. Renders call options slightly less valuable (fewer dividends captured) and put options slightly more valuable.

The Greeks

Delta (Δ) Rate of change of option price with respect to the underlying price:

$$\Delta_{\text{call}} = e^{-qT}\mathcal{N}(d_1), \qquad \Delta_{\text{put}} = -e^{-qT}\mathcal{N}(-d_1)$$

Delta ranges from 0 to e^{-qT} for calls and $-e^{-qT}$ to 0 for puts. An ATM call has $\Delta \approx 0.5$. Used for delta hedging: a delta-neutral portfolio has no first-order sensitivity to spot moves. The heatmap shows how Delta varies across the entire strike–expiry space simultaneously.

Gamma (Γ) Rate of change of Delta with respect to the underlying price; second derivative of option price with respect to S:

$$\Gamma = \frac{e^{-qT}\mathcal{N}'(d_1)}{S\sigma\sqrt{T}}$$

Always non-negative for long options. Highest for ATM options near expiry — these require the most frequent rebalancing to maintain a delta hedge. Gamma and Theta are in direct opposition: long Gamma benefits from large moves but costs Theta (time decay) to carry.

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Theta (Θ) Rate of change of option price with respect to time (time decay), expressed per calendar day. For a call with dividends:

$$\Theta_{\text{call}} = \frac{-Se^{-qT}\mathcal{N}'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}\mathcal{N}(d_2) + qSe^{-qT}\mathcal{N}(d_1)$$

Almost always negative for long options — the option loses value as it approaches expiry. The exception is deep ITM puts. Theta is the "rent" paid for holding Gamma.

Vega (ν) Sensitivity of option price to a one-percentage-point change in implied volatility (expressed per 1% move):

$$\text{Vega} = \frac{Se^{-qT}\mathcal{N}'(d_1)\sqrt{T}}{100}$$

Always positive for long options — rising IV increases the value of any option. Maximised for ATM options with long time to expiry. Note: Vega is not a Greek letter but a star name; it was adopted in practice for its phonetic similarity to the Greek letter ν.

Rho (ρ) Sensitivity of option price to a one-percentage-point change in the risk-free interest rate:

$$\rho_{\text{call}} = \frac{KTe^{-rT}\mathcal{N}(d_2)}{100}, \qquad \rho_{\text{put}} = \frac{-KTe^{-rT}\mathcal{N}(-d_2)}{100}$$

Calls have positive Rho (higher rates increase call value by increasing the forward price). Puts have negative Rho. The least impactful Greek for short-dated equity options but material for long-dated structures and interest rate derivatives.

Call Option A derivative contract granting the buyer the right, but not the obligation, to purchase the underlying asset at the strike price K on or before expiry T. The payoff at expiry is $\max(S_T - K,\ 0)$.

Put Option A derivative contract granting the buyer the right, but not the obligation, to sell the underlying asset at the strike price K on or before expiry T. The payoff at expiry is $\max(K - S_T,\ 0)$.

European Option An option exercisable only at expiry, not before. BSM prices European options in closed form. The alternative is an American option, which may be exercised at any time before expiry and generally requires numerical methods to price.

Heatmap A two-dimensional grid visualisation where cell colour encodes the magnitude of the variable of interest (here, a Greek) at each (strike, expiry) coordinate. Allows the practitioner to identify regions of high sensitivity across the full parameter space simultaneously.

Isoline / Contour Line A curve on the heatmap connecting all points sharing the same value of the displayed Greek. Analogous to elevation contours on a topographic map. Twelve isoline levels are rendered in the terminal to reveal the geometric structure of the surface.

Panel 03 — Correlation Matrix Analyser

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Asset Universe

Correlation A normalised measure of linear co-movement between two random variables X and Y:

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, +1]$$

$\rho = +1$ implies perfect positive co-movement. $\rho = -1$ implies perfect negative co-movement. $\rho = 0$ implies no linear relationship. In portfolio risk, correlations determine diversification benefits: low or negative correlations reduce portfolio variance.

Correlation Matrix A symmetric $n \times n$ matrix $\mathbf{R}$ where entry $R_{ij} = \rho(r_i, r_j)$ for assets i and j. The diagonal is identically 1. A valid correlation matrix must be positive semi-definite. The terminal models a 10-asset universe: SPX, NDX, TLT, GLD, CRU, BTC, EUR/USD, USD/JPY, HY CDX, VIX.

The asset universe is defined as follows:

Ticker	Description	Asset Class
SPX	S&P 500 Index	Equity
NDX	Nasdaq-100 Index	Equity (Tech)
TLT	20+ Year US Treasury ETF	Rates
GLD	Gold ETF	Commodity
CRU	Crude Oil (WTI)	Commodity
BTC	Bitcoin	Digital Asset
EUR/USD	Euro vs US Dollar	FX
USD/JPY	US Dollar vs Japanese Yen	FX (Safe Haven)
HY CDX	High Yield Credit Default Swap Index	Credit
VIX	CBOE Volatility Index	Volatility

Market Regimes

Market Regime A distinct state of the financial system characterised by a stable pattern of returns, volatilities, and correlations. Correlations are non-stationary: they shift dramatically between regimes and tend to spike during crises, a phenomenon known as *correlation contagion*.

Normal / Low Vol Regime A calm market environment characterised by low realised volatility (VIX typically below 20), positive equity returns, and correlations at their "structural" levels. Diversification benefits are greatest in this regime.

Stress / Crisis Regime A risk-off environment (e.g., 2008 Global Financial Crisis, March 2020 COVID crash). Correlations between all risky assets (SPX, NDX, HY CDX, BTC, CRU) spike dramatically toward +1 as indiscriminate de-risking dominates. Simultaneously, correlations with safe-haven assets (TLT, JPY) spike toward −1. This "correlation contagion" is the primary cause of diversified portfolio failures in crises: everything falls together precisely when the investor needs diversification most.

Post-Crisis Recovery The period following a crisis peak in which risk assets rebound as

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systemic risk recedes. Correlations begin normalising but remain elevated relative to the pre-crisis regime. Risk-on positioning returns gradually.

QE / Risk-On Regime A quantitative easing environment in which central bank asset purchases suppress volatility and broadly inflate asset prices. Correlations between risk assets are elevated because all are responding to the same monetary policy impulse. Effective rank is reduced.

Estimation

Ledoit-Wolf Shrinkage Estimator A regularisation technique for covariance matrix estimation introduced by Ledoit & Wolf (2004). The sample covariance matrix $\hat{\Sigma}$ is an unbiased but noisy estimator when the number of observations n is not much larger than the number of assets p. Shrinkage improves the estimator by pulling it toward a structured target $\mathbf{T}$:

$$\hat{\Sigma}_{\text{LW}} = (1-\lambda)\hat{\Sigma} + \lambda\mathbf{T}$$

In this implementation $\mathbf{T}$ is the identity matrix (scaled by asset variances), representing the prior that all assets are uncorrelated. Shrinkage reduces estimation error in finite samples at the cost of introducing a small bias.

Shrinkage Parameter (λ) Controls the degree of regularisation. $\lambda = 0$: pure sample matrix (no regularisation). $\lambda = 1$: full shrinkage toward the identity (maximum regularisation). In empirical work, $\lambda \in [0.1, 0.3]$ is typical for monthly equity return data.

Estimation Noise (σ_ε) Simulated additive noise applied to the correlation matrix entries to represent the finite-sample estimation uncertainty that would be present when estimating correlations from a limited historical window. Models the fact that observed correlations are always noisy estimates of a true but unobservable underlying correlation structure.

Effective Rank A measure of the number of statistically independent risk factors driving a portfolio. Computed here as:

$$\text{Effective Rank} = \frac{N}{1 + (N-1)\bar{\rho}^2}$$

where $\bar{\rho}$ is the average off-diagonal correlation and N is the number of assets. In a stress regime, effective rank collapses toward 1 — all assets behave as a single factor. In a normal regime, effective rank is close to N, reflecting genuine diversification.

Panel 04 — Monte Carlo Simulation Engine

Stochastic Processes

Monte Carlo Simulation A computational method for estimating the statistical properties of a complex system by generating a large number of random realisations of a probabilistic model and observing the empirical distribution of outcomes. In quantitative finance, Monte Carlo is used for pricing path-dependent derivatives, computing risk measures, and stress testing portfolios. Accuracy improves as $O(1/\sqrt{N})$ with the number of paths N.

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GBM — Geometric Brownian Motion The stochastic process underlying the Black-Scholes model. Under the physical measure:

$$\mathrm{d}S_t = \mu S_t \, \mathrm{d}t + \sigma S_t \, \mathrm{d}W_t$$

where W_t is a standard Brownian motion. The exact discretisation over a time step Δt is:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\, Z\right], \quad Z \sim \mathcal{N}(0,1)$$

GBM produces lognormally distributed prices, preserves positivity, and has independent increments. Its main empirical failure is that equity returns exhibit excess kurtosis (fat tails) and negative skewness not captured by the lognormal distribution.

Drift (μ) The annualised expected return of the asset. The deterministic component of GBM controlling the average direction of price evolution. In a risk-neutral pricing context, μ is replaced by the risk-free rate r. In physical measure simulations (such as this one), μ reflects the expected total return.

Volatility (σ) The annualised standard deviation of log-returns. The stochastic scaling factor in GBM. Controls the width of the distribution of outcomes: higher σ produces wider percentile bands and a broader terminal distribution.

Brownian Motion / Wiener Process (W_t) A continuous-time stochastic process satisfying: $W_0 = 0$; increments $W_t - W_s \sim \mathcal{N}(0, t-s)$ for $t > s$; non-overlapping increments are independent; paths are continuous but nowhere differentiable. Serves as the fundamental source of randomness in all diffusion-based models.

Merton Jump-Diffusion An extension of GBM by Robert Merton (1976) incorporating a Poisson jump process to capture sudden, discontinuous price movements (crashes, earnings surprises, geopolitical shocks). The dynamics are:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\, Z\right] + J_t$$

where the jump term $J_t = -\xi S_t(1 + \mathcal{N}(0, 0.3) \cdot \epsilon)$ with probability $\lambda\Delta t$, and $J_t = 0$ otherwise. Jump-diffusion models produce fatter tails and greater negative skewness than pure GBM, better matching the empirical distribution of equity returns.

Poisson Jump Process A counting process N_t in which jumps arrive randomly and independently at a constant average rate λ (events per unit time). The probability of exactly k jumps in interval $[0, t]$ is:

$$\mathbb{P}(N_t = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

In the terminal, each time step checks whether a jump occurs by comparing $U \sim \mathcal{U}(0,1)$ against the probability $\lambda\Delta t$.

Jump Intensity (λ) The expected number of jumps per year. $\lambda = 0.05$ implies approximately one jump every 20 years on average. Higher λ increases the frequency of discrete price discontinuities. The expected number of jumps over horizon T is λT.

Jump Size (ξ) The expected magnitude of a price decline when a jump occurs, expressed as a percentage of the current price. Jumps in this model are directionally negative (downward) to capture crash risk. The actual jump size is randomised: $J = -\xi S_t(1 + \epsilon)$ where $\epsilon \sim \mathcal{N}(0, 0.3)$.

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CEV — Constant Elasticity of Variance A model (Cox, 1975) where the local volatility of the asset depends on the price level:

$$\mathrm{d}S = \mu S \ \mathrm{d}t + \sigma_0 S^{\beta} \ \mathrm{d}W$$

When $\beta < 1$, volatility increases as the stock price falls — consistent with the *leverage effect* documented by Black (1976): as a firm's equity value falls, its leverage increases, raising the volatility of equity returns. The β parameter is fixed at 0.7 in the terminal.

Variance Reduction & Statistics

Antithetic Variates A variance reduction technique. For every simulated path driven by random numbers $\{Z_1, Z_2, \ldots, Z_T\}$, a paired "antithetic" path is generated using $\{-Z_1, -Z_2, \ldots, -Z_T\}$. Since the antithetic path is negatively correlated with the original, their average eliminates odd moments of the estimation error, reducing variance at no additional computational cost. Implemented for even-indexed paths in the terminal.

Percentile Bands Lines connecting the p-th percentile of the cross-sectional distribution of simulated prices at each time step t. The terminal renders five bands: 5th, 25th, 50th (median), 75th, and 95th percentiles. Together they summarise the full distribution of outcomes over time, revealing both central tendency and tail risk.

$E[S_T]$ — Expected Terminal Value The sample mean of all simulated terminal prices: $\widehat{E[S_T]} = \frac{1}{N}\sum_{i=1}^{N} S_T^{(i)}$. Under GBM, the theoretical value is $E[S_T] = S_0 e^{\mu T}$.

$P(S_T > S_0)$ — Probability of Profit The fraction of simulated paths whose terminal value exceeds the initial price $S_0 = 100$:

$$\hat{p} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{S_T^{(i)} > S_0\}$$

Under GBM, the theoretical probability of profit is $\mathcal{N}\left(\frac{(\mu - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right)$.

CVaR at 5% — Conditional Value at Risk Also called *Expected Shortfall* (ES). The expected loss given that the loss exceeds the 5% Value at Risk threshold:

$$\mathrm{CVaR}_{5\%} = -E\left[S_T \,\middle|\, S_T \leq \mathrm{VaR}_{5\%}\right]$$

More informative than VaR alone because it captures the severity of losses in the tail, not merely their threshold. A cornerstone of Basel III regulatory capital requirements and used universally in institutional risk management. In the terminal, CVaR is computed as the sample average of the bottom 5% of simulated terminal prices.

Terminal Distribution The probability distribution of S_T (asset price at the final time step T) implied by the simulation. Under pure GBM, this is lognormal. Under Merton jump-diffusion, it exhibits negative skewness and excess kurtosis (fat left tail). Under CEV with $\beta < 1$, it is skewed to the left due to the leverage effect.

Trading Days / Δt 252 trading days is the standard market convention for the number of business days in a calendar year. The time step is $\Delta t = 1/252$, meaning each simulation step represents exactly one trading day. This convention is used universally for annualising volatilities and scaling daily P&L.

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Mathematical Notation Reference

Symbol	Meaning
S	Spot price of the underlying asset
K	Strike price
T	Time to expiry (years)
r	Continuously compounded risk-free rate
q	Continuously compounded dividend yield
σ	Implied or realised volatility (annualised)
μ	Drift / expected return (annualised)
F	Forward price: $F = Se^{(r-q)T}$
k	Log-moneyness: $k = \ln(K/F)$
d_1, d_2	BSM intermediate variables
$\mathcal{N}(x)$	Standard normal CDF
$\mathcal{N}'(x)$	Standard normal PDF
Δ	Delta: $\partial C/\partial S$
Γ	Gamma: $\partial^2 C/\partial S^2$
Θ	Theta: $\partial C/\partial t$
ν	Vega: $\partial C/\partial \sigma$
ρ	Rho: $\partial C/\partial r$
λ	Jump intensity (jumps/year) or shrinkage parameter
ξ	Jump size
W_t	Standard Brownian motion
$\mathrm{d}t$	Infinitesimal time increment
Z	Standard normal random variate: $Z \sim \mathcal{N}(0,1)$
Δt	Discrete time step ($= 1/252$)
$\hat{\Sigma}$	Sample covariance matrix
$\mathbf{R}$	Correlation matrix
$\bar{\rho}$	Average off-diagonal correlation
N	Number of Monte Carlo paths
S_0	Initial asset price ($=$ 100 in the terminal)

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